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Reg No.

Time : 3.00 hrs

Quarterly Examination - 2025

PHYSICS

Max. Marks : 70

PART - A

15×1=15

Answer all the questions. Multiple choice questions.

1. A parallel plate capacitor stores a charge Q at a voltage V. Suppose the area of the parallel plate capacitor and the distance between the plates are each doubled then which is the quantity that wi change?
   a) Capacitance b) Charge c) Voltage d) Energy density
2. Two metallic spheres of radii 1cm and 3cm are given charges of -1×10^{-2}C and 5×10^{-2}C respectively. these are connected by a conducting wire, the final charge on the biggest sphere is
   a) 3×10^{-2}C b) 4×10^{-2}C c) 1×10^{-2}C d) 2×10^{-2}C
3. An electric dipole moment p is aligned parallel to a uniform electric field E. The energy required rotate the dipole by 90° is
   a) pE^2 b) p^2E c) pE d) infinity
4. The internal resistance of a 2.1V cell which gives a current of 0.2A through a resistance of 10Ω is
   a) 0.2 Ω b) 0.5 Ω c) 0.8 Ω d) 1.0 Ω
5. In Joule's heating law, when R and t are constant, if H is taken along y-axis and I along the x-axis, graph is
   a) parabola b) straight line c) ellipse d) circle
6. A thin insulated wire forms a plane spiral of N=100 tight turns carrying a current =8mA. The rad inside and outside turns are a=50mm and b=100mm respectively. The magnetic induction at the ce of the spiral is
   a) 5 μT b) 7 μF c) 8 μT d) 10 μT
7. A non conducting charged ring carrying a charge at q, mass m and radius r is rotated about its axi constant angular speed w. Find the ratio of its magnetic moment with angular momentum.
   a) $\frac{3q}{2m}$ b) $\frac{2q}{m}$ c) $\frac{q}{2m}$ d) $\frac{5q}{4m}$
8. The potential energy of magnetic dipole whose dipole moment is $\vec{P_m} = (-0.5\,\hat{i} + 0.4\,\hat{j})$ Am² kept in u magnetic field $\vec{B} = -0.2\,\hat{i}$ T ..... a) -0.1J b) -0.8J c) 0.1J d) 0.8J
9. In circular coil, when number of turns is doubled and resistance becomes twice that of the initial, then inductance becomes
   a) 4 times b) 2 times c) 8 times d) No change
10. A step-down transformer reduces the supply voltage from 220V to 11V and increase the current from 6A to 100A. Then its efficiency is
    a) 1.2 b) 0.83 c) 0.12, d) 0.9
11. An inductor 20mH, a capacitor 50 μF and a resistor 40Ω are connected in series across a so emf v=10 sin 340t. The power loss in AC circuit is
    a) 0.76W b) 0.89W c) 0.46W d) 0.67W
12. Which of the following is not true for electromagnetic waves?
    a) transports energy b) it transports momentum c) it transports angular momentum

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d) in vacuum, it travels with different speeds which depends on their frequency

13. Which of the following electromagnetic radiations is used for viewing objects through fog
a) microwave b) gamma rays c) X-rays d) infrared

14. A convex mirror has a focal length f. A real object is placed at a distance 'f' in front of it from the pole produces an image at a) f/2 b) infinity c) f d) 2f

15. Stars twinkle due to a) reflection b) total internal reflection c) refraction d) polarization

PART - B

Answer any six questions. Q.No.24 is compulsory. 6x2=12

16. Define: Electric field.
17. What is corona discharge?
18. If the resistance of coil is 3Ω at 20°C and $\alpha = 0.004/°C$ then determine its resistance at 100°C.
19. State Fleming's Left Hand rule.
20. What are the methods to increase the current sensitivity of a galvanometer?
21. State Lenz's Law.
22. What is Fraunhofer lines?
23. The sky appear blue in day time. Why?
24. The angle of minimum deviation for an equilateral prism is 37°. Find the refractive index of the material of the prism.

PART - C

Answer any six questions. Q.No.33 is compulsory. 6x3=18

25. What are the differences between coulomb force and gravitational force?
26. State: Kirchoff's Laws.
27. Explain: Principle of Potentio meter
28. How can galvanometer can be converted in an ammeter? Explain.
29. Explain various power losses in a transformer.
30. List the properties of electromagnetic wave.
31. What are the advantages and disadvantages of AC over DC.
32. What is optical path and derive the equation?
33. Dielectric strength of air is $3 \times 10^6 vm^{-1}$. Suppose the radius of a hollow sphere in the Van de Graff generator is R=0.5m. Calculate the maximum potential difference created by this Van de Graft generator.

PART-D

Write the answer for all questions. 5 x 5 = 25

34. a) Derive an expression for electrostatic potential due to an electric dipole. Discuss special cases.
(OR) b) Discuss the working of cyclotron in detail.

35. a) What is Gauss law and to derive the expression for electric field due to an charged infinite plane sheet. **(OR)** b) Explain: Wheatstone bridge.

36. a) Deduce the relation for the magnetic field at a point due to an infinitely long straight conductor carrying current using Biot-Savart law.
(OR)
b) How will you induce an emf by changing relative orientation of the coil with the magnetic field?

37. a) Derive the expression for phase angle between the applied voltage and current in a series RLC Circuit. **(OR)** b) Write down Maxwell equations in integral form.

38. a) Describe the Fizeau's method to determine the speed of light. **(OR)**
b) Derive the mirror equation and the equation for lateral magnification.

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